

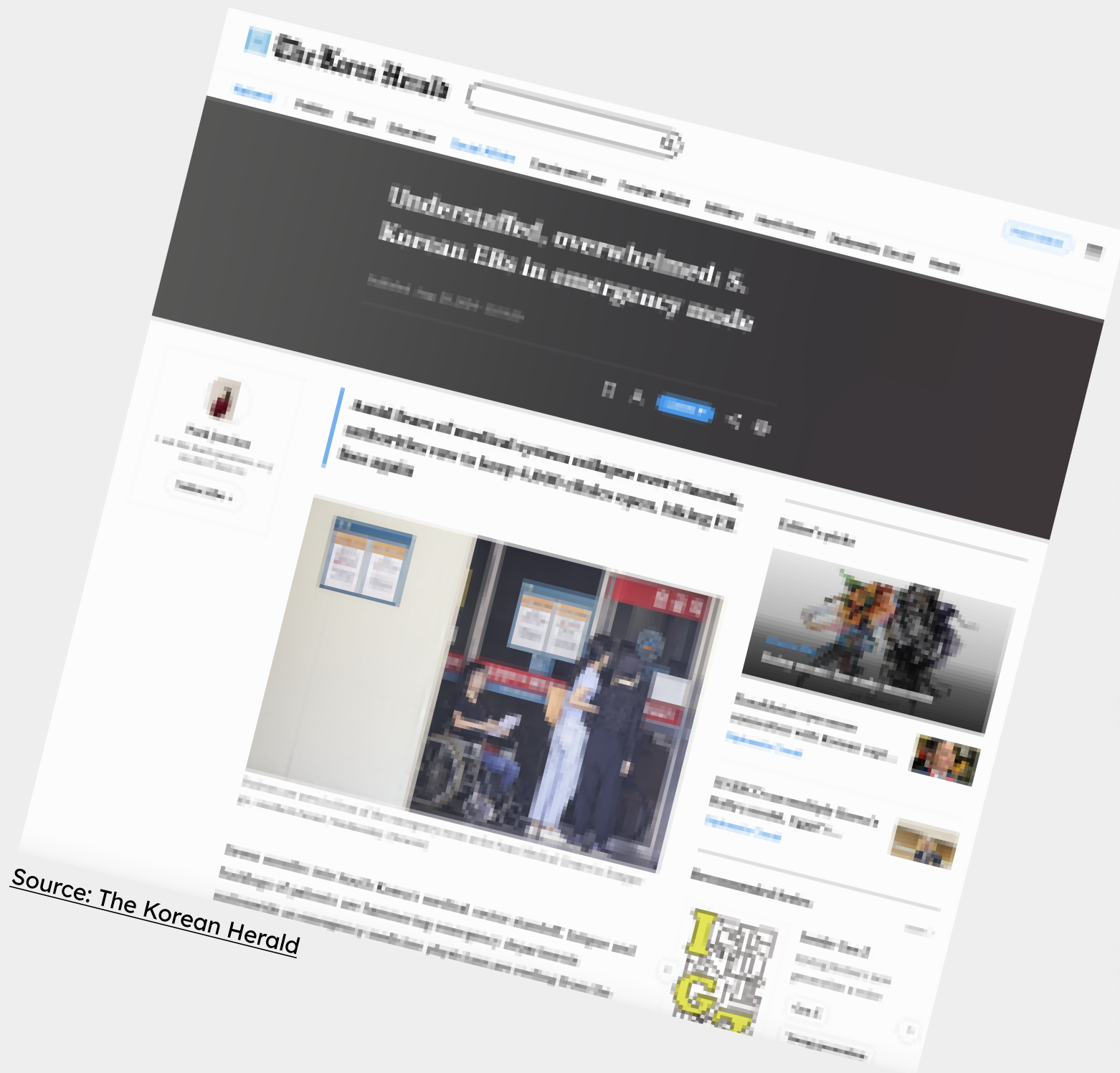


MediRoute

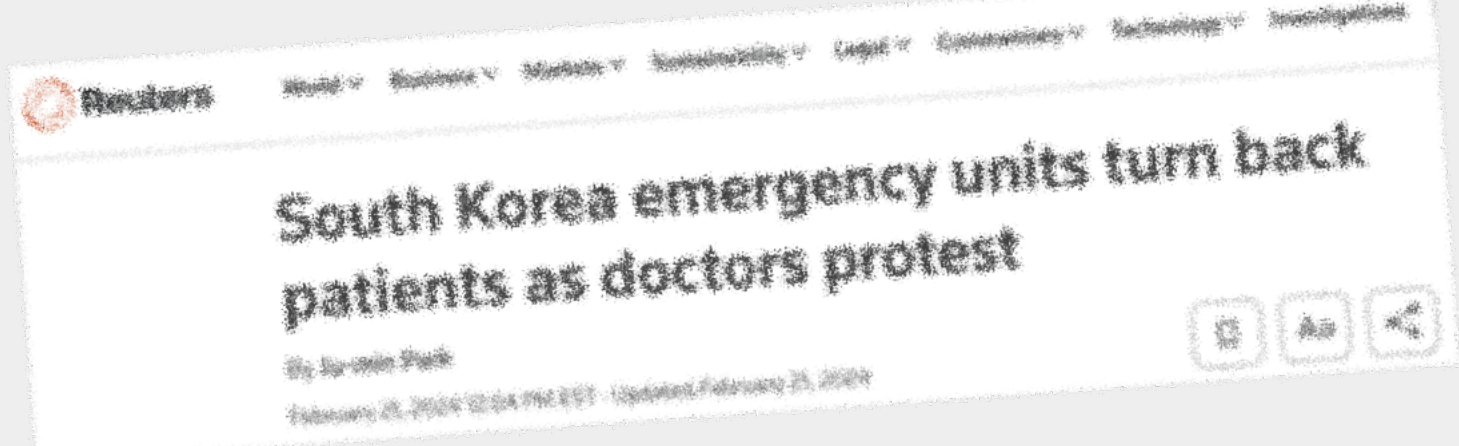
Angela Han, Ed Cho, Suhwoo Lee, Jun Bae, Jiung Moon

In an emergency, every second counts.





Source: The Korean Herald



Source: Reuters



Source: The Chosun Daily

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South Korea denies hospital emergency rooms collapsing as army doctors deployed

By Jack Kim
September 2, 2024 4:31 AM EDT · Updated September 2, 2024



Patients wait for medical treatment at Incheon Medical Center in Incheon, South Korea, April 23, 2024. REUTERS/Kim Soo-hyeon/File Photo Purchase Licensing Rights

SEOUL, Sept 2 (Reuters) - South Korea's health ministry said on Monday it was deploying military doctors to assist in some hospital emergency rooms due to a shortage of medical staff, but disputed a warning by some physicians that the system was on the verge of collapse.

A strike by young doctors has increased strain on the medical system, but Vice Health Minister Park Min-

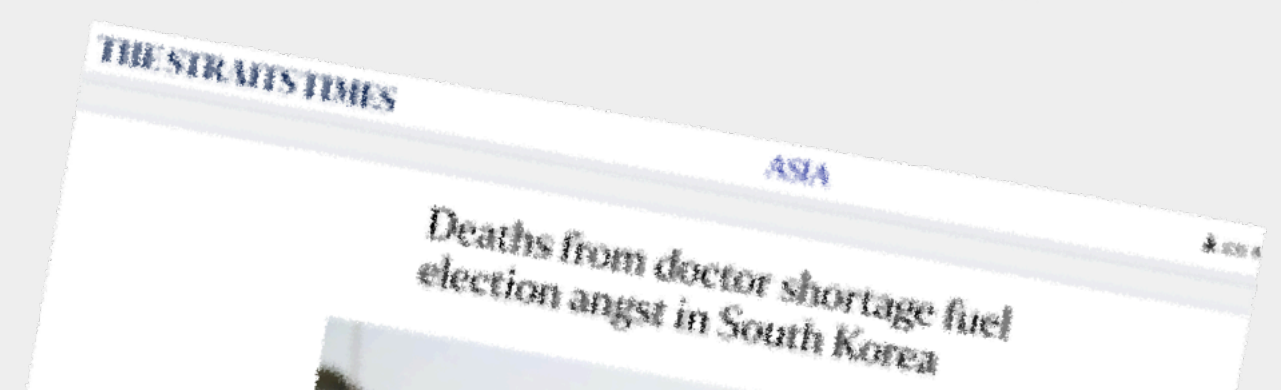


Source: The Korea Times



Source: Asia News Network

South Korea's emergency rooms are overwhelmed and understaffed, forcing some to turn patients away. Ongoing doctor shortages and strikes are putting urgent care access at serious risk.



Source: Reuters



Source: KBS News

“The paramedic union is demanding a stronger
medical info sharing system.”

Who are our target users



Name: Taemin Kwak
Role: Patient
Age: 33
Symptom: Cerebral hemorrhage

Needs

Needs automatic
routing to an
available hospital
bed

Needs real-time
transmission of
condition to
responders

Wants to avoid
wasted time from
hospital rejections

Situation

Unable to
communicate

Critically time-
sensitive condition

Likely To Be Alone
At Onset

Who are our target users



Name: Dr. Elena Park
Role: Emergency Medicine Physician
Age: 33
Workplace: Seoul National Hospital

Needs

Real-Time Bed
Availability

Smart Hospital
Recommendations
Based on
Symptoms

Situation

Patient is
unresponsive

Patient seems to
have possible
stroke

Welcome to MediRoute

Optimize patient transport with real-time hospital data and AI-driven insights.



Sign in

Create an account

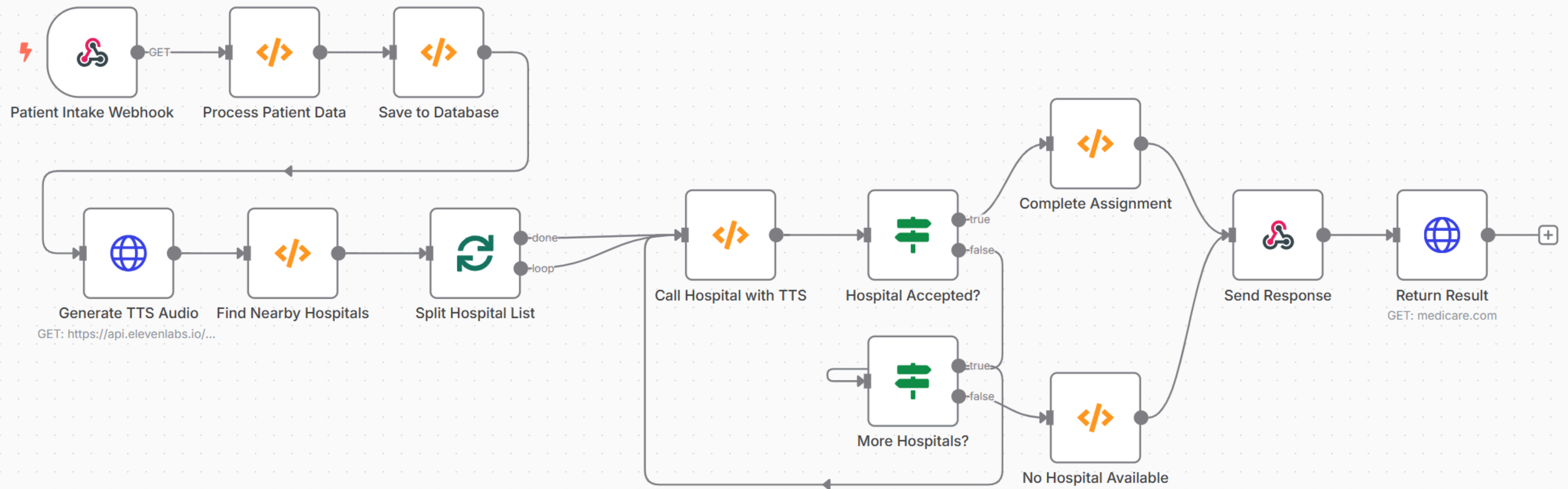
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MediRoute: AI-powered emergency navigation platform that connects patients to the right hospital based on their real-time condition and hospital capacity.

By combining symptom analysis with **live bed availability** and **TTS alerts**, it helps save time and lives.

Demo Time!



Business Model

B2C Revenue Model

Monthly subscription: \$4.99
(After 2years)

one-time payment \$6.49
with coin recharging system

Target Users

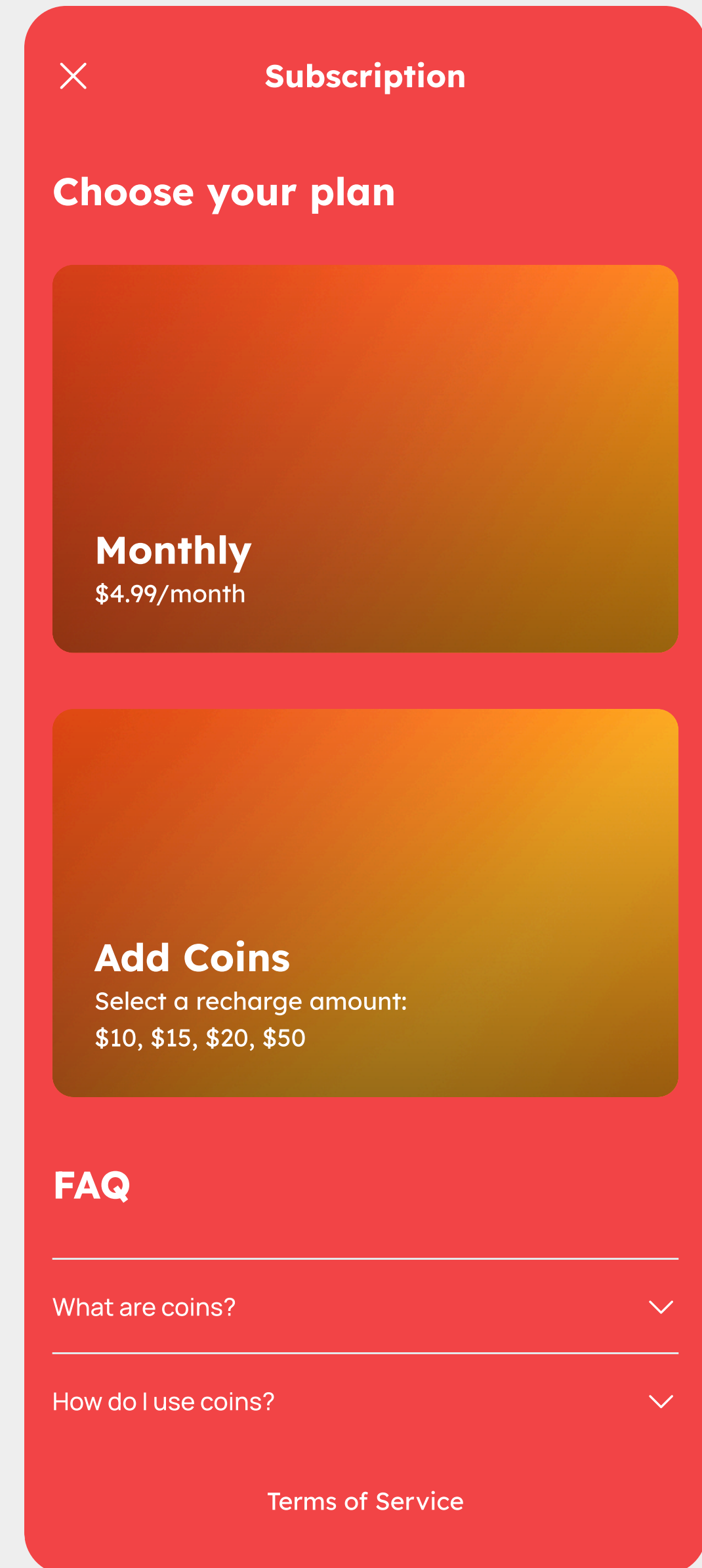
- Adults aged 65 and older
- High-risk diabetic patients
- Patients with uncontrolled or severe hypertension
(Basically, most emergency patients)

Value Proposition

- Instantly locate available ER beds
- Share patient condition via TTS before arrival
- Reduce fatal delays from hospital rejection

Channels

- iOS / Android mobile app
- Emergency medical content partnerships (YouTube, blog, Naver, etc.)



Go To Market Strategy

Year 1-2	Free, trust-building phase through hospital & community partnerships
Year 3	Monetization via \$2.99/month subscription with AI triage & hospital matching
Core Goal	Build medical trust before monetization — then convert through urgency & added value

Total Addressable Market (TAM)

Metric	Value
Annual Emergency Room Visits	14.65 million people
ARPU (Average Revenue Per User)	\$40/year
Subscription Price	\$4.99/month (adjusted conservatively)
Applicable Audience Ratio	100% (TAM, based on total emergency room visitors)

Total: 14.65M * \$40 = **\$586,000,000/year**

Serviceable Obtainable Market (SOM)

Category	Details
Annual Emergency Room Visits	14.65 million cases/year
Potential Mobile App Users	30% of ER patients
Targetable Patients	~4.4 million users/year
ARPU (Average Revenue Per User)	~40/year
B2C SAM Estimate	~176 million/year

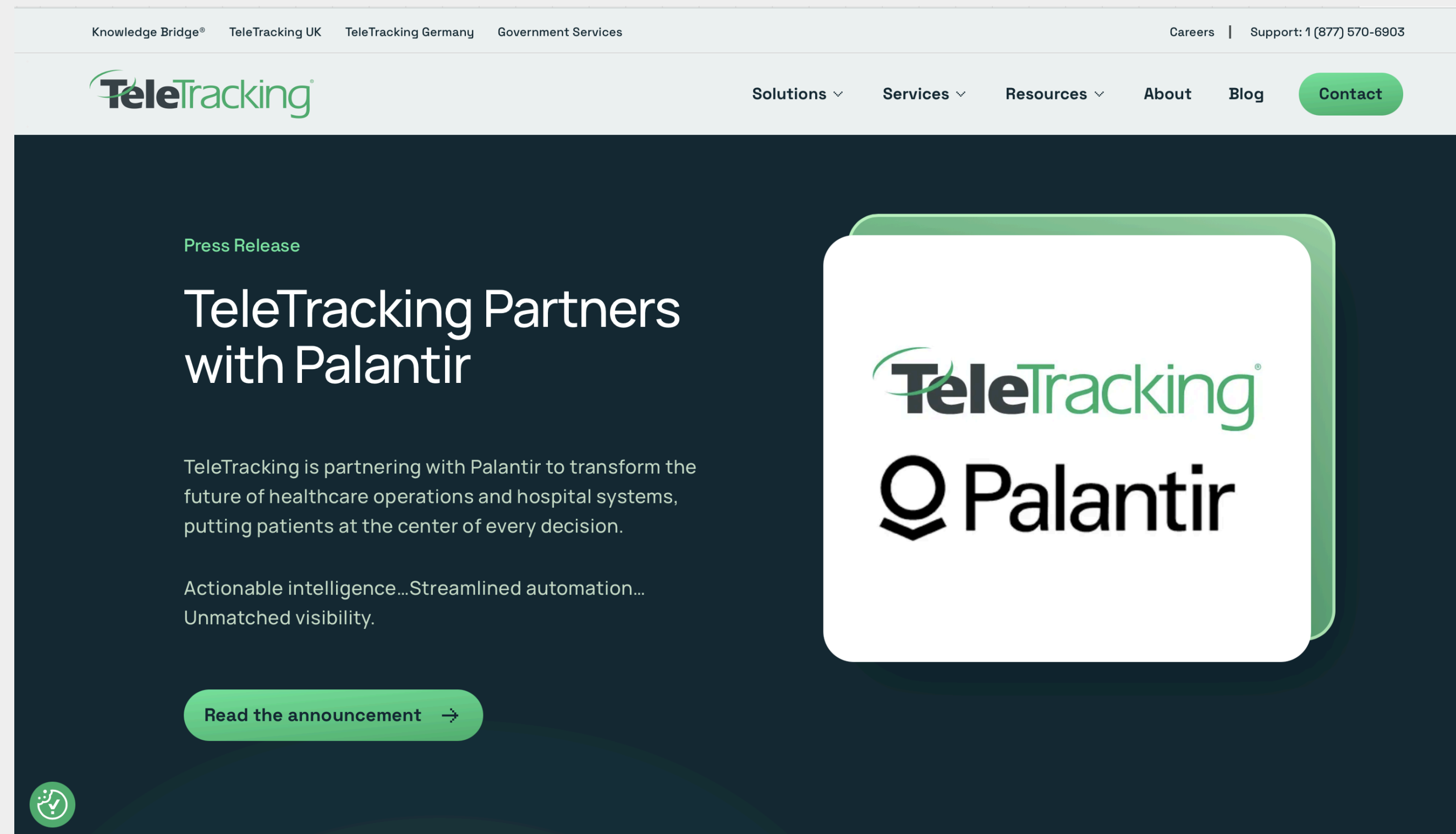
Total: 4.4M * \$40 = **\$176M/year**

Serviceable Available Market (SAM)

Metric	Value
Geographic Focused on Seoul	45%
Initial Reach	5% of Seoul focused rate
Conversion Rate (to subscription model)	20% of initial reach
ARPU (Average Revenue Per User)	\$40/year

Total: $4.4\text{M} * 45\% * 5\% * 20\% * \$40 = \text{\$7.9M Dollars/year}$

Competitors



Not in Korea, however. **TeleTracking** is the **U.S. market** leader in dedicated hospital capacity and flow management. It's an enterprise-level platform focused on managing inpatient flow and bed usage within hospitals.

It seems similar. However...

Feature	TeleTracking	MediRoute
Primary Users	Hospital operations staff	Paramedics, dispatchers, local health officials
Bed Data Usage	Internal bed management	External hospital availability check
Patient Condition Tracking	Not included	Used for matching & prioritization
Communication	Dashboard-based	Real-time TTS alerts
Environment	On-premise hospital system	Cloud-based, mobile friendly
Target Institutions	Large hospitals	Hospitals, EMS, local governments



- Each year, approximately **11 million patients** visit emergency rooms in South Korea.
 - Of these, around **10% (1.1 million patients)** are classified as critical cases (e.g., cardiac arrest, stroke, hemorrhage, severe trauma).
 - Patients are often transported to hospitals that cannot admit them, resulting in re-transfers and long wait times.
 - In some regions, bed assignment delays reach **30 minutes** to over an hour, which can be life-threatening.
- For conditions like stroke, cardiac arrest, or myocardial infarction, **every 10-minute delay in treatment may reduce the survival rate by 3–10%.**
 - If MediRoute shortens the average delay by even 10 minutes, survival outcomes can improve significantly.
 - If MediRoute improves survival by just 3% for the estimated 1.1 million critical patients annually → Over **33,000 lives** could be saved every year.
- This number is same as Harvard student population.

Our application will be a leading platform which will save people as many as possible.